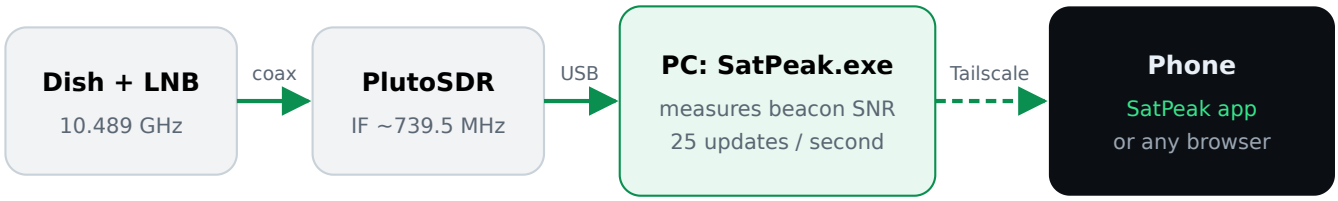
SatPeak

QO-100 dish peaking meter: how-to guide



The PC does the measuring; the phone is a live remote display you carry to the dish.

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SatPeak 1.0.0 · Windows installer `SatPeak-Setup-1.0.0.exe` · Android app `SatPeak.apk` · October 2026

1 What you need

- **Dish and LNB** pointed roughly at QO-100 (25.9° E). The LNB needs power from a bias-tee or inserter, because the Pluto cannot supply it.
- **ADALM-Pluto (PlutoSDR)** connected to the PC by USB, with the LNB output on its RX port.
- **Windows 10/11 PC** (64-bit) with **Tailscale** installed and signed in.
- **A phone** with Tailscale signed in to the same account. On Android, use the SatPeak app. On iPhone or anything else, use the web browser.

> Only one program can use the Pluto at a time. Close SDR Console, SATSAGEN, Linkbox, SDR++ and similar programs before starting SatPeak.

2 Install on the PC

1. Double-click **SatPeak-Setup-1.0.0.exe**. You don't need administrator rights. Click **Next**, tick *Create a desktop shortcut* if you want one, then click **Install**.
2. Leave **Start SatPeak now** ticked and click **Finish**. A black SatPeak window opens. It finds the Pluto on USB automatically.
3. **Windows Firewall:** the first time, Windows may ask whether SatPeak can use the network. Click **Allow** (Private networks is enough). If you don't, the phone can't connect.
4. Note the address shown in the window, for example:
   `100.101.102.103:8765`
   This is your PC's Tailscale address (yours will differ), and it doesn't change.

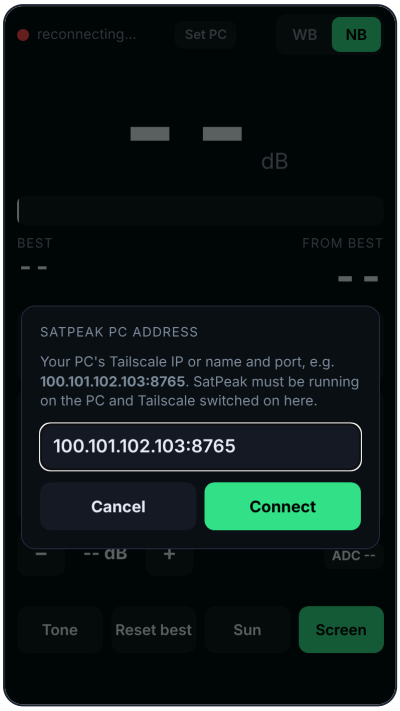
After that, start SatPeak from the **Start menu → SatPeak** or the desktop icon. Close the window to stop it. **SatPeak (demo mode)** runs with a simulated signal, which is handy for testing the phone connection indoors without the Pluto.

3 Install the phone app

Android

1. Get **SatPeak.apk** onto the phone. You can tap it in the Claude chat on your phone, email it to yourself, or copy it over USB. On the PC it's in `Desktop\SatPeak` and in *Start menu → SatPeak Android app (APK)*.
2. Tap the file. If Android says installs from this source are blocked, tap **Settings**, turn on **Allow from this source**, go back and tap **Install**. If Play Protect warns about an unknown developer, tap **Install anyway**. The app is signed by you, not by Google Play.
3. Turn **Tailscale on** on the phone.

4. Open **SatPeak**. The address box appears the first time. Enter your PC address (for example `100.101.102.103:8765`) and tap **Connect**. The app remembers it. To change it later, tap the address button at the top.



First start: enter the PC address

iPhone or any other device

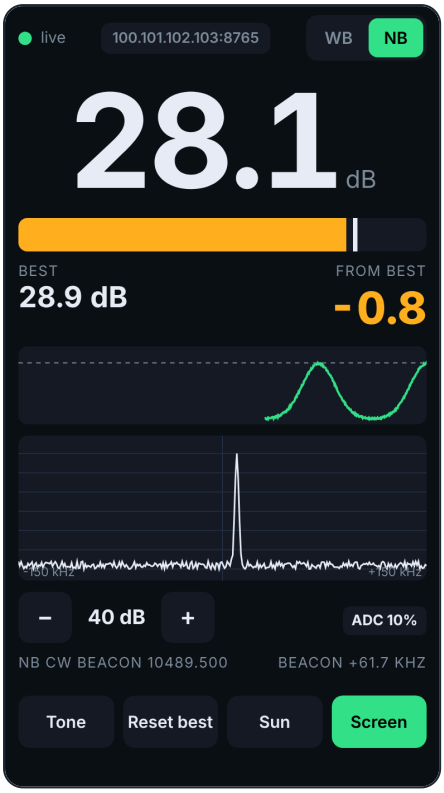
With Tailscale on, open the browser and go to:

`http://100.101.102.103:8765`

The page is the same as the app. Two differences: the screen may dim while you work, so turn up auto-lock before you go out. And the **Screen** keep-awake button needs HTTPS, which you can set up with `tailscale serve`.

> The Android app keeps the screen on automatically while it's open, so you don't need to change any phone settings.

4 The screen explained



Live reading on the NB beacon

1. **Status dot**: green means live data, with the round-trip time in ms. Amber means connected but no data; red means reconnecting. The button next to it is the PC address, which you tap to change.

2. **WB / NB** selects the beacon. **NB** is the CW beacon at 10489.500 MHz and is the default. **WB** is the DVB-S2 beacon at 10491.500 MHz, for when it's on air.

3. **Big number**: beacon signal-to-noise in dB, updated about 25 times a second. Higher is better.

4. **Bar**: the current level. The white marker shows the best level so far.

5. **From best**: how far you are below your best reading. It shows **green** within 0.5 dB, **amber** within 2 dB, and **red** beyond that. This is the number to watch.

6. **History**: the last 30 seconds. The dashed line is your best.

7. **Spectrum** around the beacon. In NB mode you should see a single sharp spike.

8. **Gain − / +** changes the Pluto receive gain in 2 dB steps. **ADC %** shows how hard the input is driven, and it reads **CLIPPING** when it's too hard.

9. **Beacon +61.7 kHz** is your LNB's frequency error. A steady value means a healthy PLL LNB.

10. **Tone** gives an audio pitch that rises with signal. **Reset best** clears the peak hold. **Sun** is a high-contrast mode for daylight. **Screen** keeps the display awake.

5 Peaking the dish, step by step

1. **At the PC:** start SatPeak and check that the window shows the address with no errors.
2. **At the dish:** open the app and check that the dot is green. Press **Reset best**.
3. **Set the gain:** use **−** / **+** until the ADC stays well below 90% and never shows CLIPPING. About 40 dB is typical.
4. **Turn on Tone** so you can peak by ear without looking at the screen.
5. **Azimuth:** slowly swing the dish left and right through the peak. Watch the big number and the history trace, and stop where *From best* is green (±0.0). Then tighten the azimuth lightly.
6. **Elevation:** do the same up and down, and stop at the best reading.
7. **LNB skew:** rotate the LNB a few degrees each way in its holder and keep the position with the highest reading.
8. **Repeat** azimuth and elevation with small movements. Each pass should gain a little less. Stop when nothing improves by more than about 0.3 dB.
9. **Tighten everything:** watch *From best* while you do up the bolts. If it drops, loosen, re-peak and tighten again.

> **Tips.** Move slowly. The reading is averaged slightly, so give it about half a second after each move. Press **Reset best** after any big move so the peak hold doesn't mislead you. Reflections from your body can change the reading by tenths of a dB, so step away when judging the final peak.

6 Troubleshooting

Problem	What to do
Phone dot stays red ("reconnecting")	Check that Tailscale is **on** on the phone and that the SatPeak window is open on the PC. Check the address (including `:8765`). Allow SatPeak in Windows Firewall: *Windows Security* → *Firewall & network protection* → *Allow an app* → tick SatPeak.exe.
Dot is amber ("no data")	The PC can't read the Pluto. Look at the SatPeak window for an error. Close other SDR programs, then re-plug the Pluto and restart SatPeak.
Window says *Could not open the Pluto*	The Pluto isn't plugged in, or another program is using it. Try a different USB port or cable. **SatPeak (demo mode)** confirms that the rest of the system works.
Window says *Port 8765 is already in use*	SatPeak is already running in another window. Close the extra one.
Reading is about 0 dB in WB mode	The wideband beacon may be off air. Use **NB**. If it's also around 8 dB (noise only), check that the LNB is powered and the dish is roughly aimed.
NB shows no clear spike	Your LNB may be more than 150 kHz off. Run SatPeak with `--lo` adjusted, see section 7. Also check that the LNB has power.
ADC badge shows CLIPPING	Press **−** gain until it clears. Clipping flattens the peak and makes peaking impossible.
Reading jumps around	Wind, or the dish isn't tight. Use **History** to judge the average, or run with `--smooth 0.2` for more averaging.
Android won't install the APK	Allow installs from unknown sources for the app you opened it with (Files, Gmail, Chrome). Android 7 or later is needed. If an older SatPeak is installed with a different signature, uninstall it first.

7 Advanced options and files

SatPeak.exe accepts options. Edit the shortcut's *Target* or run it from a command prompt in `%LOCALAPPDATA%\Programs\SatPeak`.

Option	Meaning
`--demo`	Simulated signal; no Pluto needed
`--lo 9750`	LNB local oscillator in MHz (default 9750). Example: `--lo 9749.94` corrects signals that appear 60 kHz high .
`--mode nb` / `wb`	Beacon to start on (default nb)
`--gain 40`	Starting receive gain, 0–70 dB
`--uri auto`	Pluto address: auto (default), `usb:1.1.5` or `ip:192.168.2.1`
`--port 8765`	Network port for the phone
`--rate 25`	Updates per second
`--smooth 0.35`	Averaging: 1 = none (fastest), 0.1 = heavy

Where things are

Installed program	`%LOCALAPPDATA%\Programs\SatPeak`. Uninstall from *Settings → Apps* or the Start menu.
Installer, APK, source	`Desktop\SatPeak`. Rebuild the installer with `build\build_exe.bat`.
Android signing key	`build\android\satpeak-release.p12`, alias `satpeak`, password `satpeak-qo100`. **Keep it safe**: updates to the phone app must be signed with the same key.

> **Example:** on the author's station the NB CW beacon was received at IF 739.5625 MHz, so signals appear about 62 kHz high (LNB LO about 9749.94 MHz). That's normal for a PLL LNB and inside SatPeak's ±150 kHz search.